# Contents

# 1 Measurable Functions

## 1.1 Elementray Properties of Measruable Functions

**Definition 1.1.1**

We endow $\overline{\mathbb{R}}$ with a topology called the order topology in the following manner. For each $a \in \mathbb{R}$ let

$$L_a = \overline{\mathbb{R}} \cap \{x : x < a\} = [-\infty, a) \quad \text{and} \quad R_a = \overline{\mathbb{R}} \cap \{x : x > a\} = (a, \infty].$$

The collection $\mathcal{S} = \{L_a : a \in \mathbb{R}\} \cup \{R_a : a \in \mathbb{R}\}$ is taken as a subbasis for this topology. A basis for the topology is given by

$$\mathcal{S} \cup \{R_a \cap L_b : a, b \in \mathbb{R}, a < b\}.$$

Observe that the topology on $\mathbb{R}$ induced by the order topology on $\overline{\mathbb{R}}$ is precisely the usual topology on $\mathbb{R}$.

**Definition 1.1.2 ▶ Measurable Functions**

- Suppose $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ are measure spaces. A mapping $f : X \to Y$ is called ***measurable with respect to $\mathcal{M}$ and $\mathcal{N}$*** if $f^{-1}(E) \in \mathcal{M}$ whenever $E \in \mathcal{N}$. If there is no danger of confusion, reference to $\mathcal{M}$ and $\mathcal{N}$ will be omitted, and we will simply use the term "measurable mapping."
- If $Y$ is a topological space, a restriction is placed on $\mathcal{N}$. In this case it is always assumed that $\mathcal{N}$ is the $\sigma$-algebra of Borel sets $\mathcal{B}$. Thus, in this situation, a mapping $(X, \mathcal{M}) \xrightarrow{f} (Y, \mathcal{B})$ is measurable if

$$f^{-1}(E) \in \mathcal{M} \text{ whenever } E \in \mathcal{B}.$$

- When $Y$ is taken as $\overline{\mathbb{R}}$ (endowed with the order topology) and $(X, \mathcal{M})$ is a topological space with $\mathcal{M}$ the collection of Borel sets. Then $f$ is called a ***Borel measurable function***.

- When $X = \mathbb{R}^n$, $\mathcal{M}$ is the class of Lebesgue measurable sets and $Y = \overline{\mathbb{R}}$. Here, the measurability of $f$ required that $f^{-1}(E)$ be Lebesgue measurable whenever $E \subset \overline{\mathbb{R}}$ is Borel, in which case $f$ is called a ***Lebesgue measurable function***. The definitions imply that $E$ is a measurable set if and only if $\chi_E$ is a measurable function.

**Theorem 1.1.3**

If the mapping $(X, \mathcal{M}) \xrightarrow{f} (Y, \mathcal{B})$ is measurable, where $\mathcal{B}$ is the $\sigma$-algebra of Borel sets. Define

$$\Sigma = \{E : E \subset Y \text{ and } f^{-1}(E) \in \mathcal{M}\}.$$

Then $\Sigma$ is a $\sigma$-algebra.

*Proof.* Easy to verify. □

**Corollary 1.1.4**

A continuous mapping is a Borel measurable function.

*Proof.* If $f$ is continuous, $\Sigma$ contains all open open sets. Since $\Sigma$ is a $\sigma$-algebra, $\Sigma$ contains all Borel sets. □

**Definition 1.1.5**

If $f : X \to \mathbb{R}$, we call the sets

$$\{f > a\} := \{x \in X : f(x) > a\}$$

the ***superlevel sets of $f$***.

**Theorem 1.1.6**

Let $f : X \to \overline{\mathbb{R}}$, where $(X, \mathcal{M})$ is a measure space. The following conditions are equivalent:

(i) $f$ is measurable.
(ii) $\{f > a\} \in \mathcal{M}$ for each $a \in \mathbb{R}$.
(iii) $\{f \geq a\} \in \mathcal{M}$ for each $a \in \mathbb{R}$.
(iv) $\{f < a\} \in \mathcal{M}$ for each $a \in \mathbb{R}$.

(v) $\{f \leq a\} \in \mathcal{M}$ for each $a \in \mathbb{R}$.

**Theorem 1.1.7**

A function $f : X \to \overline{\mathbb{R}}$ is measurable if and only if
(i) $f^{-1}\{-\infty\} \in \mathcal{M}$ and $f^{-1}\{\infty\} \in \mathcal{M}$ and
(ii) $f^{-1}(a, b) \in \mathcal{M}$ for all open intervals $(a, b) \subset \mathbb{R}$.

**Lemma 1.1.8**

If $f$ and $g$ are measurable functions, then the following sets are measurable:
(i) $X \cap \{x : f(x) > g(x)\}$,
(ii) $X \cap \{x : f(x) \geq g(x)\}$,
(iii) $X \cap \{x : f(x) = g(x)\}$.

*Proof.* If $f(x) > g(x)$, then there is a rational number $r$ such that $f(x) > r > g(x)$. Therefore, it follows that

$$\{f > g\} = \bigcup_{r \in \mathbb{Q}} (\{f > r\} \cap \{g < r\}),$$

which is a measurable set , and easily (i) holds. (ii) is just the complements of (i) with $f$ and $g$ interchanged. (iii) is just the intersection two of two measruable sets of type (ii), and so it too is measurable. □

**Definition 1.1.9**

If $f$ and $g$ are real-valued measurable functions, then $f + g$ is undefined at points where it would be of the form $\infty - \infty$. This difficulty is overcome if we define

$$(f + g)(x) := \begin{cases} f(x) + g(x), & x \in X - B, \\ \alpha, & x \in B, \end{cases}$$

where $\alpha \in \overline{\mathbb{R}}$ is chosen arbitrarily and where

$$B := (f^{-1}\{\infty\} \cap g^{-1}\{-\infty\}) \cup (f^{-1}\{-\infty\} \cap g^{-1}\{\infty\}).$$

**Theorem 1.1.10**

If $f, g : X \to \overline{\mathbb{R}}$ are measurable functions, then $f + g$ and $fg$ are measurable.

*Proof.* We will treat the case that $f$ and $g$ have values in $\mathbb{R}$. The proof is similar in the general case and is left as an exercise (see Exercise 2, Section 5.1).

To prove that the sum is measurable, define $F : X \to \mathbb{R} \times \mathbb{R}$ by

$$F(x) = (f(x), g(x))$$

and $G : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ by

$$G(x, y) = x + y.$$

Then $G \circ F(x) = f(x) + g(x)$, so it suffices to show that $G \circ F$ is measurable. Referring to Theorem 1.1.7, we need only show that $(G \circ F)^{-1}(J) \in \mathcal{M}$ whenever $J \subset \mathbb{R}$ is an open interval. Now $U := G^{-1}(J)$ is an open set in $\mathbb{R}^2$, since $G$ is continuous. Furthermore, $U$ is the union of a countable family, $\mathcal{F}$, of 2-dimensional intervals $I$ of the form $I = I_1 \times I_2$, where $I_1$ and $I_2$ are open intervals in $\mathbb{R}$. Since

$$F^{-1}(I) = f^{-1}(I_1) \cap g^{-1}(I_2),$$

we have

$$F^{-1}(U) = F^{-1}\left(\bigcup_{I \in \mathcal{F}} I\right) = \bigcup_{I \in \mathcal{F}} F^{-1}(I),$$

which is a measurable set. Thus $G \circ F$ is measurable, since

$$(G \circ F)^{-1}(J) = F^{-1}(U).$$

The product is measurable by essentially the same proof. $\square$

**Definition 1.1.11 ▸ Cantor Function**

Recall that the Cantor set $C$ can be expressed as

$$C = \bigcap_{j=1}^{\infty} C_j$$

Where $C_j$ is the union of the $2^j$ closed intervals that remain after the $j$th step of the construction. The set

$$D_j = [0,1] - C_j$$

consists of the $2^{j-1}$ open intervals that are deleted at the $j$th step. Let there intervals be denoted by $I_{j,k}$, $k = 1, 2, \cdots, 2^{j-1}$, and order them in the obvious way from left to right. Now define a continuous function $f_j$ on $[0,1]$ by

$$\begin{aligned} f_j(0) &= 0, \\ f_j(1) &= 1, \\ f_j(x) &= \frac{k}{2^j}, \quad x \in I_{j,k} \end{aligned}$$

and define $f_j$ linearly on each interavl of $C_j$. The function $f_j$ is continuous and nondecreasing, and it satisfies

$$|f_j(x) - f_{j+1}(x)| < \frac{1}{2^j}, \quad x \in [0,1]$$

Since

$$|f_j - f_{j+m}| < \sum_{i=j}^{j+m-1} \frac{1}{2^i} < \frac{1}{2^{j-1}}$$

it follows that the sequence $\{f_j\}$ converges uniformly to a continuous function $f$, called ***Cantor-Lebesgue function***.

*Note.*
$f_j$ 在 $I_{j,k}$ 上的设定是端点的平均值.

**Corollary 1.1.12**

The Cantor-Lebesgue function $f$ and its associate $h(x) := f(x) + x$ described above have the following properties:

(i) $f(C) = [0,1]$; that is, $f$ maps a set of measure 0 onto a set of positive measure.
(ii) $h$ maps a Lebesgue measurable set onto a nonmeasurable set.
(iii) The composition of Lebesgue measurable functions need not be Lebesgue measurable.

(iv) Lebesgue measurable function dose not preserve Borel sets.

*Proof.* 1. $f : [0,1] \to [0,1]$ is onto. It is easy to see that $f(C) = [0,1]$, because $f(C)$ is compact and $f([0,1] - C)$ is countable.

2.

*Proof sketch.*
把 $C$ 看成是集合 $[0,1]$ 测度为零的"骨". $[0,1] \setminus C$ 是 $[0,1]$ 的测度为 1 的"肉". $f$ 的构造是对"骨"的展开: $f(C) = [0,1]$; 同时对"肉"的压缩: 令 $f$ 在 $I_{j,k}$ 上取常值. 通过令 $h : [0,1] \to [0,2]$. $h(x) = f(x) + x$, 我们变换映射的尺度, 不再将"肉"压缩, 而是忠实地搬运, 将每个 $I_{j,k}$ 映到大小相同的一个区间. 从而间接的看到这两个操作对"骨"的影响区别不大 (测度都是 1).

*Note.*
$h(C)$ 和 $f(C)$ 测度相同在直觉上也是存在依据的.
Cantor 函数 $f$ 是一列分段线性函数 $f_n$ 的一致极限. 在构成 Cantor 集第 $n$ 步近似 $C_n$ 的那些区间上, $f_n$ 的斜率为 $(3/2)^n$, 这是一个趋向于无穷大的值.
函数 $h(x) = f(x) + x$ 可以看作是 $h_n(x) = f_n(x) + x$ 的极限. 在同样的区间上, $h_n$ 的斜率变为 $(3/2)^n + 1$.
为一个趋于无穷的量加上 1, 并不会显著改变它的数量级. 两个斜率的比值为 $\frac{(3/2)^n+1}{(3/2)^n} = 1 + (2/3)^n \to 1$.
这便解释了我们的直觉: "+x" 这个操作虽然修复了 $f$ 对 $C$ 补集测度的" 压缩", 但对于 $f$ 自身在 $C$ 上的" 测度生成" 效应来说, 其 * 相对 * 影响是微不足道的. 因此, $f$ 和 $h$ 都将测度为零的集合 $C$ " 拉伸" 成了一个测度为 1 的集合.

$h$ is trictly increasing. Thus , h is a homeomorphism from $[0,1]$ onto $[0,2]$. Then $h$ carries the open set $[0,1] \setminus C$ to an open set. And we have

$$\begin{aligned} m(h([0,1] \setminus C)) &= m\left(h\left(\bigcup_{j,k} I_{j,k}\right)\right) \\ &= \sum_{j,k} m(h(I_{j,k})) \\ &= \sum_{j,k} m(I_{j,k}) \\ &= 1 \end{aligned}$$

Therefore, $h$ maps the Cantor set to a set $P$ with measure 1. From Exercise **??** , there exists a non-measurable set $N \subseteq P$. Set $A = h^{-1}(N)$. Since $h^{-1}(N) \subseteq h^{-1}(P) = C$, $A$ is a subset of measure-zero set $C$. Thus $A$ is Lebesgue-measurable. But $h(A) = N$ is not measurable.

3. Note that $h^{-1}$ is measurable, since it is continuous. Let $F := h^{-1}$. Observe that $\chi_A$ is measurable, but
$$\chi_A \circ F = \chi_A \circ h^{-1} = \chi_N$$
is not Lebesgue-measurable.

4. Observe that $A$ is not a Borel set, for if it were, then $F^{-1}(A)$ would be a Borel set. But $F^{-1}(A) = h(A) = N$ and $N$ is not a Borel set. Now $g \coloneqq \chi_A$ is a Lebesgue measurable function such that $g^{-1}(\{1\}) = A$, where $\{1\}$ is a Borel sets but $A$ is not.

$\square$

**Theorem 1.1.13**

Suppose $f : X \to \overline{\mathbb{R}}$ is measurable and $g : \overline{\mathbb{R}} \to \overline{\mathbb{R}}$ is Borel measurable. Then $g \circ f$ is measurable. In particular, if $X = \mathbb{R}^n$ and $f$ is Lebesgue measurable, then $g \circ f$ is Lebesgue measurable.

**Corollary 1.1.14**

Let $f : X \to \overline{\mathbb{R}}$ be a measurable function.

(i) Let $\varphi(x) = |f(x)|^p$, $0 < p < \infty$, and let $\varphi$ assume arbitrary extended values on the sets $f^{-1}(\infty)$ and $f^{-1}(-\infty)$. Then $\varphi$ is measurable.

(ii) Let $\varphi(x) = \frac{1}{f(x)}$, and let $\varphi$ assume arbitrary extended values on the sets $f^{-1}(0)$, $f^{-1}(\infty)$ and $f^{-1}(-\infty)$. Then $\varphi$ is measurable.

In particular, if $X = \mathbb{R}^n$ and $f$ is Lebesgue measurable, then $\varphi$ is Lebesgue measurable in (i) and (ii).

*Proof.* 1. Define $g(t) = |t|^p$ for $t \in \mathbb{R}$, and assign arbitrary caluse to $g(\infty)$ and $g(-\infty)$. Now apply the prvious theorem.

2. Proceed in a similar way by defining $g(t) = \frac{1}{t}$ when $t \neq 0, \infty, -\infty$ and assigning arebitrary values to $g(0)$, $g(\infty)$, and $g(-\infty)$ .

$\square$

### Theorem 1.1.15

Let $(X, \mathcal{M}, \mu)$ be a complete measure space and let $f, g$ be extended real-valued functions defined on $X$. If $f$ is measurable and $f = g$ almost everywhere, then $g$ is measurable.

*Remark.*
In a complete measure space, we can consider a function $f$ that just define on almost everywhere on $X$, by extending $f$ into $\overline{f}$ with arbitrary value taken on where $f$ was not defined. .

*Proof.* Let $N = \{x : f(x) = g(x)\}$. Then $\mu(\tilde{N}) = 0$, and thus $\tilde{N}$ as well as all subsets of $\tilde{N}$ are measurable. For $a \in \mathbb{R}$, we have

$$\begin{aligned}\{g > a\} &= (\{g > a\} \cap N) \cup (\{g > a\} \cap \tilde{N}) \\ &= (\{f > a\} \cap N) \cup (\{g > a\} \cap \tilde{N}) \in \mathcal{M}.\end{aligned}$$

□

### Theorem 1.1.16

Suppose $\varphi$ is an outer measure on a space $X$. Then an extended real-valued function $f$ on $X$ is $\varphi$-measurable if and only if

$$\varphi(A) \geq \varphi(A \cap \{f \leq a\}) + \varphi(A \cap \{f \geq b\})$$

whenever $A \subset X$ and $a < b$ are real numbers.

*Proof.* The "only if" is immediate from

$$\varphi(A) \geq \varphi(A \cap \{f \leq a\}) + \varphi(A \cap \{f > a\}) \geq \varphi(A \cap \{f \leq a\}) + \varphi(A \cap \{f \geq b\})$$

To show the converse, we need to prove that for any $r \in \mathbb{R}$, the set

$$E \coloneqq \{f \leq r\}$$

satisfies

$$\varphi(A) \geq \varphi(A \cap E) + \varphi(A - E)$$

for each $A \subseteq X$. We define

$$B_j \coloneqq A \cap \left\{r + \frac{1}{j+1} \leq f \leq r + \frac{1}{j}\right\}$$

We shall show

$$\infty > \varphi(A) \geq \varphi\left(\bigcup_{j=1}^{\infty} B_{2j}\right) = \sum_{j=1}^{\infty} \varphi\left(B_{2j}\right)$$

by induction. We fix $k$ and assume

$$\varphi\left(\bigcup_{j=1}^{k} B_{2j}\right) = \sum_{j=1}^{k} \varphi\left(B_{2j}\right)$$

We set

$$A_k \coloneqq \bigcup_{j=1}^{k} B_{2j}$$

Then

$$\left(A_k \cup B_{2(k+1)}\right) \cap \left\{f \leq r + \frac{1}{2(k+1)}\right\} = B_{2(k+1)}$$

$$\left(A_k \cup B_{2(k+1)}\right) \cap \left\{f \geq r + \frac{1}{2k+1}\right\} = A_k$$

We have

$$\varphi\left(\bigcup_{j=1}^{k+1} B_{2j}\right) = \varphi\left(A_k \cup B_{2(k+1)}\right) \geq \varphi\left(B_{2(k+1)}\right) + \varphi(A_k) = \sum_{j=1}^{k+1} \varphi\left(B_{2j}\right)$$

Thus we inductively show that

$$\varphi(A) \geq \varphi\left(\bigcup_{j=1}^{\infty} B_{2j}\right) = \sum_{j=1}^{\infty} \varphi\left(B_{2j}\right)$$

Similarly, one can show that

$$\varphi(A) \geq \varphi\left(\bigcup_{j=1}^{\infty} B_{2j-1}\right) = \sum_{j=1}^{\infty} \varphi\left(B_{2j-1}\right)$$

Then the series

$$\sum_{j=1}^{\infty} \varphi\left(B_j\right) \leq 2\varphi(A) < \infty$$

convergents. The tail of the series can be arbitrarily small. For each $\varepsilon > 0$, there exists $m \in \mathbb{N}$ such that

$$\varepsilon > \sum_{j=m}^{\infty} \varphi\left(B_j\right) \geq \varphi\left(\bigcup_{j=m}^{\infty} B_j\right)$$

Where

$$\bigcup_{j=m}^{\infty} B_j = A \cap \left\{r < f \leq r + \frac{1}{m}\right\}$$

We define a measure $\psi$

$$\psi(F) := \varphi(A \cap F), \quad \forall F \subseteq X$$

Then

$$\varepsilon > \psi\left(\left\{r < f \leq r + \frac{1}{m}\right\}\right)$$

Thus

$$\begin{aligned}
\varphi(A \cap E) + \varphi(A - E) &= \psi(E) + \psi(E^c) \\
&= \psi(\{f \leq r\}) + \psi(\{f > r\}) \\
&\leq \psi(\{f \leq r\}) + \psi\left(\left\{f > r + \frac{1}{m}\right\}\right) + \psi\left(\left\{r < f \leq r + \frac{1}{m}\right\}\right) \\
&\leq \psi(\{f \leq r\}) + \psi\left(\left\{f \geq r + \frac{1}{m}\right\}\right) + \varepsilon \\
&= \varphi(A \cap \{f \leq r\}) + \varphi\left(A \cap \left\{f \geq r + \frac{1}{m}\right\}\right) + \varepsilon \\
&\leq \varphi(A) + \varepsilon
\end{aligned}$$

Since $\varepsilon$ is arbitrary, we have

$$\varphi(A \cap E) + \varphi(A - E) \leq \varphi(A)$$

Thus $E$ is measurable. $\square$

# Exercises

**Exercise 1.1.17**

Prove that a function defined on $\mathbb{R}^n$ that is continuous everywhere except for a set of Lebesgue measure zero is a Lebesgue measurable function. In particular, conclude that a nondecreasing function defined on $[0, 1]$ is Lebesgue measurable.

*Proof.* If $f$ is such function. Let $N$ be the measure-zero set where $f$ is not continuous. $f$ is continuous on the topological subspace $\mathbb{R}^n \setminus N$, thus is measurable on $(\mathbb{R}^n \setminus N, \mathcal{B}(\mathbb{R}^n \setminus N))$. Then for each open sets $U \subseteq \mathbb{R}$, we have

$$\begin{aligned} f^{-1}(U) &= \left(f^{-1}(U) \cap (\mathbb{R}^n \setminus N)\right) \cup \left(f^{-1}(U) \cap N\right) \\ &= \left(f|_{\mathbb{R}^n \setminus N}^{-1}(U)\right) \cap \left(f^{-1}(U) \cup N\right) \end{aligned}$$

Where $f|_{\mathbb{R}^n \setminus N}^{-1}(U) \in \mathcal{B}(\mathbb{R}^n \setminus N)$. Note that

$$\mathcal{B}(\mathbb{R}^n \setminus N) = \{B \cap (\mathbb{R}^n \setminus N) : B \in \mathcal{B}(\mathbb{R}^n)\} \subseteq \mathcal{L}(n)$$

Since $f^{-1}(U) \cap N \in \mathcal{L}(n)$, we have $f^{-1}(U) \in \mathcal{L}(n)$. $f$ is Lebesuge-measurable.

Since a nondecreasing function is continuous except for a countable set, which is Lebesgue-measura-zero. It is Lebesgue measurable from our argument above. □

## 1.2 Limits of Measurable Functions

**Definition 1.2.1**

Let $(X, \mathcal{M}, \mu)$ be a measure space, and let $\{f_i\}$ be a sequence of measurable functions defined on $X$. The ***upper*** and ***lower envelopes*** of $\{f_i\}$ are defined respectively as

$$\sup_i f_i(x) = \sup\{f_i(x) : i = 1, 2, \dots\}$$

and

$$\inf_i f_i(x) = \inf\{f_i(x) : i = 1, 2, \dots\}.$$

Also, the ***upper*** and ***lower limits*** of $\{f_i\}$ are defined as

$$\limsup_{i\to\infty} f_i(x) = \inf_{j\geq 1}\left(\sup_{i\geq j} f_i(x)\right)$$

and

$$\liminf_{i\to\infty} f_i(x) = \sup_{j\geq 1}\left(\inf_{i\geq j} f_i(x)\right).$$

**Theorem 1.2.2**

Let $\{f_i\}$ be a sequence of measurable functions defined on the measure space $(X, \mathcal{M}, \mu)$. Then $\sup_i f_i$, $\inf_i f_i$, $\limsup_{i\to\infty} f_i$, and $\liminf_{i\to\infty} f_i$ are all measurable functions.

*Proof.* For each $a \in \mathbb{R}$ the identity

$$X \cap \{x : \sup_i f_i(x) > a\} = \bigcup_{i=1}^{\infty}(X \cap \{f_i(x) > a\})$$

implies that $\sup_i f_i$ is measurable. The measurability of the lower envelope follows from

$$\inf_i f_i(x) = -\sup_i(-f_i(x)).$$

Now that it has been shown that the upper and lower envelopes are measurable, it is immediate that the upper and lower limits of $\{f_i\}$ are also measurable. □

**Theorem 1.2.3 ▸ Egorov**

Let $(X, \mathcal{M}, \mu)$ be a finite measure space and suppose $\{f_i\}$ and $f$ are measurable functions that are finite almost everywhere on $X$. Also, suppose that $\{f_i\}$ converges pointwise a.e. to $f$. Then for each $\varepsilon > 0$ there exists a set $A \in \mathcal{M}$ such that $\mu(A^c) < \varepsilon$ and $\{f_i\} \to f$ uniformly on $A$.

*Remark.*
定义一个依赖于 $x \in X$ 和两个自然数索引 $n, k \in \mathbb{Z}_+$ 的逻辑陈述 $P_{n,k}(x)$, 它的值为 1 或 0, 代表真与假.

- $k$ 表示精度, $k$ 越大条件越严格.

- $n$ 表示阶段或步骤

我们可以定义

- 逐点条件:

$$\forall k \in \mathbb{Z}_+, \forall x \in E, \exists N \in \mathbb{Z}_+, \forall n \geq N, P_{n,k}(x) = 1$$

即对于任意的精度 $k$ , 在每个点上可以最终保持的.
设 $E_{n,k} = P_{n,k}^{-1}(1)$, 即该精度在某一阶段成立的点. 那么逐点条件用集合, 就是

$$E \subseteq \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} E_{n,k} = \liminf_{n\to\infty} E_{n,k}, \quad \forall k \in \mathbb{Z}_+$$

- 一致条件:

$$\forall k \in \mathbb{Z}_{>0}, \exists N_k \in \mathbb{Z}_{>0}, \forall n \geq N_k, \forall x \in E, P_{n,k}(x) = 1$$

即对于任意的精度 $k$, 最终总是全局保持的.

Egorov 定理就是说, 如果 $P_{n,k}(x)$ 在每个点总是最终精确的, 那么它差不多总是最终整体精确的.

*Proof.* Let $E$ be the set on which the function $f_i$, $i = 1, 2, \cdots$ are defined and finite. Let $F$ be the set on which $f$ convergent pointwise to $f$. Let $A_0 = E \cap F$. We have by hypothesis $\mu(A_0^c) = 0$. For each $\varepsilon > 0$, define

$$E_{i,k} = \left\{ x \in A_0 : |f_i - f| < \frac{1}{k} \right\}$$

Define

$$A_{i,k} = \bigcap_{j=i}^{\infty} E_{j,k} = \left\{ x \in A_0 : \left| f_j(x) - f(x) \right| < \frac{1}{k}, \forall j \geq i \right\}$$

Then $A_{1,k} \subseteq A_{2,k} \subseteq \cdots$, $A_0 = \bigcup_{i=1}^{\infty} A_{i,k}, \forall k$. $A_0 \setminus A_{1,k} \supseteq A_0 \setminus A_{2,k}, \cdots$. $\bigcap_{i=1}^{\infty} \left( A_0 \setminus A_{i,k} \right) = \varnothing$.

$$0 = \mu(\varnothing) = \lim_{i\to\infty} \mu\left(A_0 \setminus A_{i,k}\right)$$

Then for each $k$ , and for each $\varepsilon_k > 0$, there exists $i_k$ such that

$$\mu\left(A_0 \setminus A_{i_k,k}\right) < \varepsilon_k$$

*Remark.*
这里就是在说, 处处最终成立的条件, 是差不多最终一直成立的.

$$E \subseteq \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} P_n \implies E_{\text{大测度, 舍弃了不多于}\varepsilon} \subseteq \bigcap_{n=N_\varepsilon}^{\infty} P_n$$

如果 $P_n$ 都是可测条件

For each $\varepsilon > 0$, we take $\varepsilon_k = \frac{\varepsilon}{2^k}$, let $A = \bigcap_{k=1}^{\infty} A_{i_k,k}$. Then

$$\mu(A_0 \setminus A) = \mu\left(\bigcup_{k=1}^{\infty} A_0 \setminus A_{i_k,k}\right) \leq \sum_{k=1}^{\infty} \mu\left(A_0 \setminus A_{i_k,k}\right) < \sum_{k=1}^{\infty} \varepsilon_k = \varepsilon$$

$$\mu(X \setminus A) \leq \mu(X \setminus A_0) + \mu(A_0 \setminus A) = 0 + \mu(A_0 \setminus A) < \varepsilon$$

And note that for each $k$ , since $A \subseteq A_{i_k,k}$ we have

$$\left|f_j(x) - f(x)\right| < \frac{1}{k}, \quad \forall x \in A, j \geq i_k$$

Then

$$\lim_{i\to\infty} \sup_{x\in A} |f_i - f| = 0$$

That is $\{f_i\}$ convergents uniformly to $f$ on $A$. □

**Corollary 1.2.4**

Let $\{E_n\}$ be a sequence of measurable functions. If

$$E \subseteq \liminf_{n\to\infty} E_n$$

Then for each $\varepsilon > 0$, there exists measurable set $E_\varepsilon$, and $N_\varepsilon \in \mathbb{Z}_{>0}$, such that $\mu(E_\varepsilon^c) < \varepsilon$ and

$$E_\varepsilon \subseteq \bigcap_{n=N_\varepsilon} E_n$$

*Remark.*
也就是说, 如果一列可测的条件 $P_n$ , 对于每个点 $x \in E$ 总是最终保持的. 相当于说每个 $x$ 总会在每一个阶段加入并保持在 $P$ 中, 那么我们可以让还没有加入的那

些成员任意地少，然后把它们抛弃掉，使得剩下的成员在一个共同的阶段后总是全都保持在里面的.

## Definition 1.2.5

A sequence of measurable functions $\{f_i\}$ defined relative to the measure space $(X, \mathcal{M}, \mu)$ is said to ***converge in measure*** to a measurable function $f$ if for every $\varepsilon > 0$, we have

$$\lim_{i\to\infty} \mu(X \cap \{x : |f_i(x) - f(x)| \geq \varepsilon\}) = 0.$$

## Theorem 1.2.6

Let $(X, \mathcal{M}, \mu)$ be a finite measure space, and suppose $\{f_i\}$ and $f$ are measurable functions that are finite a.e. on $X$. If $\{f_i\}$ converges to $f$ a.e. on $X$, then $\{f_i\}$ converges to $f$ in measure.

*Remark.*
测度的有限性是为了保证 Egorov 的使用，以及从一致逼近过渡到测度逼近的可行性.

## Theorem 1.2.7

Let $(X, \mathcal{M}, \mu)$ be a measure space and let $\{f_i\}$ and $f$ be measurable functions such that $f_i \to f$ in measure. Then there exists a subsequence $\{f_{i_j}\}$ such that

$$\lim_{j\to\infty} f_{i_j}(x) = f(x)$$

for $\mu$-a.e. $x \in X$.

*Remark.*
依测度收敛导致子列的 a.e. 收敛.

*Proof sketch.*
将收敛描述成某种“逃逸难度”，依测度收敛就是在任意逃逸难度（函数值差大于 $\frac{1}{k}$）下，逃逸的可能性随着进程的推进最终会趋于“不可能逃逸”(逃逸出去的点最终是零测的)．于是我们选取某些进程 $i_k$，使得在这个进程之后，即便逃逸难度很小，逃逸可能性也会很微弱．依测度收敛保证了几乎所有的点都被这些进程给捕

> 获. 于是最终几乎所有点在任意简单的难度下都不能逃逸, 也就代表了几乎处处收敛.

*Proof.* Let $i_1$ be an integer such that

$$\mu\left(\left\{x \in X : |f_{i_1}(x) - f(x)| \geq 1\right\}\right) < \frac{1}{2}$$

Assume that we take $i_1, \cdots, i_k$, then we take $i_{k+1} > i_k$ such that

$$\mu\left(\left\{x \in X : |f_{i_{k+1}}(x) - f(x)| \geq \frac{1}{k+1}\right\}\right) < \frac{1}{2^{k+1}}$$

Let

$$A_j = \bigcup_{k=j}^{\infty} \left\{x \in X : |f_{i_k} - f| \geq \frac{1}{k}\right\}$$

Then

$$A_1 \supseteq A_2 \supseteq \cdots$$

$$\mu(A_j) \leq \sum_{k=j}^{\infty} \mu\left(\left\{x \in X : |f_{i_k} - f| \geq \frac{1}{k}\right\}\right) < \sum_{k=j}^{\infty} \frac{1}{2^k} = \frac{1}{2^{j-1}}$$

Specifically, $\mu(A_1) < \infty$, define $A = \bigcap_{j=1}^{\infty} A_j$ we have

$$\mu(A) = \lim_{j \to \infty} \mu(A_j) = 0$$

For each $x \in A^c = \bigcup_{j=1}^{\infty} A_j^c$, there exists $k_0$ such that $x \in A_{k_0}^c$,

$$A_{k_0}^c = \bigcap_{k=k_0}^{\infty} \left\{x \in X : |f_{i_k} - f| < \frac{1}{k}\right\}$$

Which mean that

$$|f_{i_k}(x) - f(x)| < \frac{1}{k}, \quad \forall k \geq k_0$$

Thus $\lim_{k \to \infty} f_{i_k}(x) = f(x)$. Since $x$ is arbitrary, $\{f_{i_k}\}_k$ converges to $f$ on $A^c$. □

**Theorem 1.2.8**

- A sequence $\{f_i\}$ of a.e. finite-valued measurable functions on a measure space $(X, \mathcal{M}, \mu)$ is ***fundamental in measure*** if for every $\varepsilon > 0$,

$$\mu(\{x : |f_i(x) - f_j(x)| \geq \varepsilon\}) \to 0$$

- Prove that if $\{f_i\}$ is fundamental in measure, then there is a measurable function $f$ to which the sequence $\{f_i\}$ converges in measure. as $i$ and $j \to \infty$.

**Theorem 1.2.9**

Let $(X, \mathcal{M}, \mu)$ be a finite measure space. Suppose that $\{f_i\}_{i=1}^{\infty}$ and $f$ are measurable functions. Prove that $f_i \to f$ in measure if and only if each subsequence of $f_i$ has a subsequence that converges to $f$ $\mu$-a.e.

## 1.3 Approximation of Measurable Functions

**Theorem 1.3.1**

Let $f : X \to \overline{\mathbb{R}}$ be an arbitrary (possibly nonmeasurable) function. Then the following hold:

(i) There exists a sequence of simple functions, $\{f_i\}$, such that $f_i(x) \to f(x)$ for each $x \in X$.
(ii) If $f$ is nonnegative, the sequence can be chosen such that $f_i \uparrow f$.
(iii) If $f$ is bounded, the sequence can be chosen such that $f_i \to f$ uniformly on $X$.
(iv) If $f$ is measurable, the $f_i$ can be chosen to be measurable.

*Proof.* Step1. All for $f \geq 0$:

Assume that $f \geq 0$. For $i \in \mathbb{Z}_{>0}$, partition $[0, i)$ in to $i \cdot 2^i$ half-open intervals of the form $[\frac{k-1}{2^i}, \frac{k}{2^i}), k = 1, 2, \cdots, i \cdot 2^i$. Label these intervals $H_{i,k}$ and let

$$A_{i,k} = f^{-1}\left(H_{i,k}\right), \quad A_i = f^{-1}\left([i, \infty]\right)$$

Define $f_i$ on $X$ as

$$f_i(x) = \begin{cases} \frac{k-1}{2^i}, & x \in A_{i,k} \\ i, & x \in A_i \end{cases}$$

- If $f$ is measurable, then the sets $A_{i,k}$ and $A_i$ are measurable, and thus so are the functions $f_i$.

It is easy to verify that

$$f_1 \leq f_2 \leq \cdots \leq f$$

If $f(x) < \infty$ , then for every $i > f(x)$, we have

$$|f_i(x) - f(x)| < \frac{1}{2^i}$$

and hence $f_i(x) \to f(x)$. If $f(x) = \infty$, then $f_i(x) = i \to f(x)$. In any case, we obtains

$$\lim_{i \to \infty} f_i(x) = f(x), \quad x \in X$$

- If $f$ is bounded by $M$. Then $A_i = \varnothing$ for all $i > M$, and therefore $|f_i(x) - f(x)| < \frac{1}{2^i}$ for all $x \in X$, thus showing that $f_i \to f$ uniformly .

Step 2. The general case:

Applying to $f^+$ and $f^-$, (i), (iii), (iv) holds .

□

**Theorem 1.3.2 ▸ Lusin's theorem**

Suppose $(X, \mathcal{M}, \mu)$ is a measure space where $X$ is a metric space and $\mu$ is a finite Borel measure. Let $f : X \to \mathbb{R}$ be a measurable function that is finite almost everywhere. Then for every $\varepsilon > 0$ there is a closed set $F \subset X$ with $\mu(F) < \varepsilon$ such that $f$ is continuous on $F$ in the relative topology.